

EDEXCEL INTERNATIONAL A LEVEL

WMA12/01 January 2025 Worked Solutions

January 2025 paper rebuilt with the worked solution after each question.

10

paper questions

WMA12

Pure 2

**Single-
paper
solutions**standalone IAL
route**Dr Eslam Ahmed**Prepared for Dr Eslam Ahmed - eliteigcse.com**P2**

PAST PAPER

WMA12/01 January 2025

January 2025 | 10 questions | 75 marks

10

questions

75

marks

Answers

worked solution
after each
question

Question 1

Arithmetic Sequences

1. The arithmetic series S is given by

$$S = 2 + 5 + 8 + 11 + \dots + 254$$

Find

- (a) the number of terms in the series,

(2)

- (b) the sum of the series.

(2)

(Total for Question 1 is 4 marks)

(Total for Question 1 is 4 marks)

Worked Solution - Question 1

1. Use the arithmetic formula

Identify the first term, common difference, and required term or sum from the question.

2. Substitute and solve

Use the arithmetic term or sum formula, then simplify the requested value.

3. State the final result

Write the answer in the form requested by the question: The required value or expression is obtained by the final simplification.

Final answer

The required value or expression is obtained by the final simplification.

Question 2

Binomial Expansion

2. (a) Find the first 4 terms, in ascending powers of x , of the binomial expansion of

$$(2 - 5x)^8$$

giving each term in simplest form.

(4)

This expansion is to be used to find an approximation for 2.05^8

- (b) State the value of x that should be used.

(There is no need to carry out this calculation.)

(1)

(Total for Question 2 is 5 marks)

(Total for Question 2 is 5 marks)

Worked Solution - Question 2

1. Choose the required binomial terms

Write the expansion term-by-term and keep only the powers needed for the coefficient or approximation.

2. Compare coefficients

Collect like powers, compare with the given expression, and solve for the required constant.

3. State the final result

Write the answer in the form requested by the question: $+ - + - + - +$ Constant term 82 OR 256

Final answer

$+ - + - + - +$ Constant term 82 OR 256

Question 3

Applications of Differentiation

3.

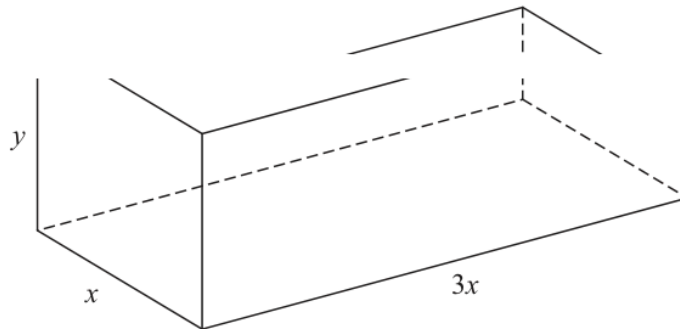


Figure 1

Figure 1 shows an open-topped container used for holding water.

The container is in the shape of a cuboid and is made of sheet metal.

The base of the container is a rectangle $3x$ metres by x metres.

The height of the container is y metres as shown in Figure 1.

Given that the capacity of the container is 120 m^3

(a) show that the area $A\text{ m}^2$ of the sheet metal used to make the container is given by

$$A = Px^2 + \frac{Q}{x}$$

where P and Q are positive constants to be found.

(4)

(b) Use calculus to find the value of x for which A has a stationary value, giving your answer to 3 significant figures.

(4)

(c) Find $\frac{d^2A}{dx^2}$ and hence show that the value of x found in part (b) gives the minimum value of A .

(2)

(Total for Question 3 is 10 marks)

(Total for Question 3 is 10 marks)

Worked Solution - Question 3

1. Differentiate and set the condition

Differentiate the expression, then use the gradient, stationary-point, tangent, normal, or optimisation condition.

2. Classify or evaluate

Substitute the required value and use the derivative information to complete the coordinate, gradient, interval, or maximum/minimum.

3. State the final result

Write the answer in the form requested by the question: = MINIMUM* *

Final answer

= MINIMUM* *

Question 4

Geometric Sequences

4. **In this question you must show all stages of your working.**
Solutions relying entirely on calculator technology are not acceptable.

Given that, in a particular geometric series,

- the sum of the first three terms is 70.2
- the sum to infinity is 75

find, for this series,

- (a) the common ratio,

(4)

- (b) the first term.

(2)

(Total for Question 4 is 6 marks)

(Total for Question 4 is 6 marks)

Worked Solution - Question 4

1. Use the geometric formula

Identify the first term, common ratio, and whether the question needs a term, finite sum, or sum to infinity.

2. Substitute and solve

Apply the correct geometric formula and simplify, checking any validity condition on the ratio.

3. State the final result

Write the answer in the form requested by the question: into 75 1 a r = - and finds a 45 a =

Final answer

into 75 1 a r = - and finds a 45 a =

Question 5

Polynomials

5.

$$f(x) = 3x^3 + ax^2 - 10x + b$$

where a and b are constants.

Given that $(3x - 4)$ is a factor of $f(x)$,

(a) show that $16a + 9b = 56$

(2)

Given further that when $f(x)$ is divided by $(x - 2)$ the remainder is b ,

(b) find the value of a and the value of b .

(4)

(c) Hence, using algebra, fully factorise $f(x)$.

(3)

(Total for Question 5 is 9 marks)

(Total for Question 5 is 9 marks)

Worked Solution - Question 5

1. Use the polynomial condition

Apply the factor theorem, remainder theorem, or substitution stated in the question to form the required equation.

2. Finish the algebra

Solve the resulting equation and substitute back where needed, then write the requested factor, coefficient, or value.

3. State the final result

Write the answer in the form requested by the question: The required result is shown.

Final answer

The required result is shown.

Question 6

6.

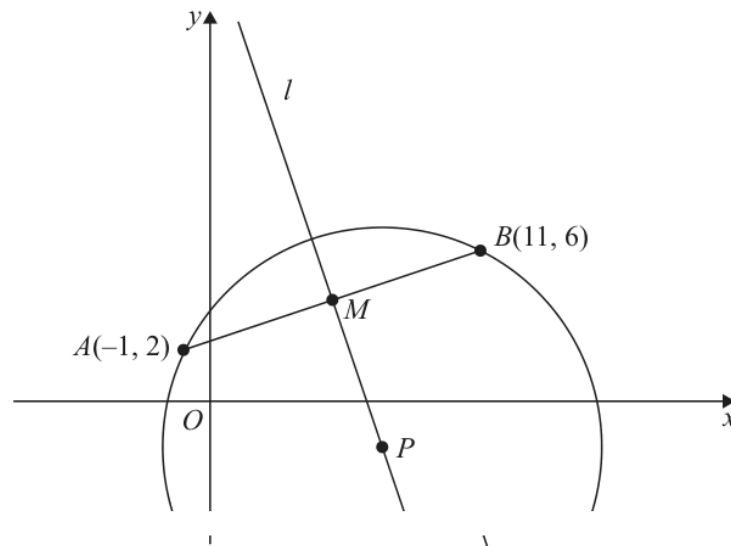


Figure 2

The point $A(-1, 2)$ and the point $B(11, 6)$ both lie on a circle with centre P .

The point M is the midpoint of AB .

Given that the line l passes through M and P , as shown in Figure 2,

- (a) find an equation for l , giving your answer in the form $y = mx + c$, where m and c are constants.

(4)

Given that P has coordinates $(7, k)$, where k is a constant,

- (b) find the value of k ,

(1)

- (c) find an equation for the circle.

(3)

(Total for Question 6 is 8 marks)

(Total for Question 6 is 8 marks)

Worked Solution - Question 6

Topic group

1. Identify the circle information

Use the centre-radius form, distance formula, gradient condition, or tangent property required by the diagram.

2. Complete the geometry

Substitute the coordinates carefully and simplify to the requested equation, length, point, or proof.

3. State the final result

Write the answer in the form requested by the question: $- + -- () ()^2 2 7 2 80 x y - + + =$

Final answer

$$- + -- () ()^2 2 7 2 80 x y - + + =$$

Question 7

Integration

7.

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

- (i) The table below shows values of x and y , where $y = \log_{10}(x + 5)$, for x values between -1 and 4

x	-1	0	1	2	3	4
$y = \log_{10}(x + 5)$	$\log_{10} 4$	$\log_{10} 5$	$\log_{10} 6$	$\log_{10} 7$	$\log_{10} 8$	$\log_{10} 9$

Using the trapezium rule with all the y values in the given table, show that

$$\int_{-1}^4 \log_{10}(x + 5) \, dx \approx \log_{10} k$$

where k is an integer to be found.

(3)

- (ii) Find the value of a such that

$$2 \log_5(5 - a) - \log_5(a + 25) = 1$$

(5)

(Total for Question 7 is 8 marks)

(Total for Question 7 is 8 marks)

Worked Solution - Question 7

1. Set up the integral

Choose the correct integrand and limits, or apply the trapezium rule if the question gives tabulated values.

2. Evaluate the area or expression

Integrate or calculate the trapezium estimate carefully, then simplify the requested value.

3. State the final result

Write the answer in the form requested by the question: The required result is shown.

Final answer

The required result is shown.

Question 8

Proof

8. (i) A student states

“If x and y are irrational numbers, $x \neq y$, then xy is also irrational.”

Show, by counter example, that this statement is not always true.

(1)

- (ii) Prove, using algebra, that for all odd integers n , the value of the expression

$$n^3 + 3n + 2$$

is always even but never a multiple of 4

(4)

(Total for Question 8 is 5 marks)

(Total for Question 8 is 5 marks)

Worked Solution - Question 8

Topic group

1. Set up the proof

Translate the statement into algebra, or choose a clear counterexample if the statement is false. Keep each implication justified.

2. Complete the argument

Simplify the algebra or evaluate the counterexample, then finish with a clear conclusion.

3. State the final result

Write the answer in the form requested by the question: proven *

Final answer

proven *

WMA12/01 JANUARY 2025

12 marks

Question 9

Integration

9.

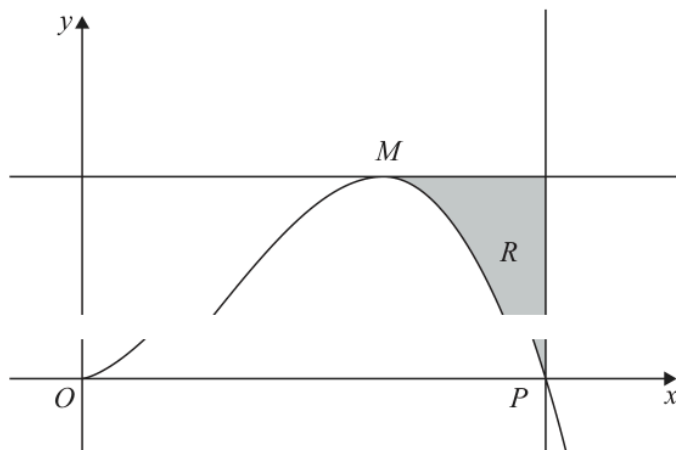


Figure 3

**In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.**

Figure 3 shows a sketch of part of the curve with equation

$$y = \frac{9x^2(5 - \sqrt{x})}{5} \quad x \geq 0$$

The curve has a turning point at the point M , as shown in Figure 3.

(a) Using calculus, find the coordinates of M .

(5)

The curve crosses the x -axis at the point P , as shown in Figure 3.

(b) Use algebra to find the x coordinate of P .

(2)

The finite region R , shown shaded in Figure 3, is bounded by the curve, the line through M parallel to the x -axis and the line through P parallel to the y -axis.

(c) Use algebraic integration to find the area of R , giving your answer to one decimal place.

(5)

(Total for Question 9 is 12 marks)

(Total for Question 9 is 12 marks)

Worked Solution - Question 9

1. Set up the integral

Choose the correct integrand and limits, or apply the trapezium rule if the question gives tabulated values.

2. Evaluate the area or expression

Integrate or calculate the trapezium estimate carefully, then simplify the requested value.

3. State the final result

Write the answer in the form requested by the question: 1312

Final answer

1312

Question 10

Trigonometric Equations

10.

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

(a) Show that

$$\cos \theta \left(3 \tan \theta + \frac{2}{\tan \theta} \right) \equiv \sin \theta + \frac{2}{\sin \theta} \quad \theta \neq \frac{n\pi}{2} \quad (4)$$

(b) Hence solve, for $0 < x < 2\pi$, the equation

$$\cos x \left(3 \tan x + \frac{2}{\tan x} \right) = 4 \sin x - 5$$

giving your answers to 3 significant figures.

(4)

(Total for Question 10 is 8 marks)

(Total for Question 10 is 8 marks)

Worked Solution - Question 10

1. Rearrange the trigonometric equation

Use the identity or ratio needed to reduce the equation to one trigonometric function in the required interval.

2. List the valid solutions

Find all angles in the given range and discard any extra values outside the interval.

3. State the final result

Write the answer in the form requested by the question: - therefore = for example

Final answer

- therefore = for example